

1.1 Classifying Parent Functions

Lesson Introduction

 Benchmark	MA.912.F.1.1 Given an equation or graph that defines a function, classify this function type. Given an input-output table, determine a function type that could represent it.			 Learning Target	I can classify families of functions from a graph or equation.
 Benchmark Coherence	Building On MA.8.F.1.2 Given a function defined by a graph or an equation, determine whether the function is a linear function. Given an input-output table, determine whether it could represent a linear function.			Working Toward MA.912.F.1.1 This benchmark begins in Algebra 1 and extends toward function types beyond the clarifications included within this benchmark toward Algebra 2.	
 Essential Questions	- How can a function be transformed based on given values? - How can I use features of a graph to classify its function type?				
 Lesson Narrative	In this lesson, students are introduced to the concept of a parent function in each of the eight function types for the Algebra 1 course. Students will engage in a gallery walk to review the function, table, and graph representations of each type. They will be introduced to transformations of graphs using one of the parent functions in this lesson.				
 Component Summary	1.1.1 Warm-Up (~5 min.)	Would You Rather? Icebreaker (SE p. 3)	1.1.3 Guided Practice (10 min.)	Review given functions and determine how each one should be classified based on the function or graph, and learn about translations and reflections of graphs (SE p. 8)	
	1.1.2 Activity (20 min.)	Gallery walk that explores the eight different types of parent functions (SE p. 4)	1.1.4 Your Turn! (5 min.)	Determine the function family for a given equation or graph (SE p. 9)	
	1.1.5 Wrap-Up (~5 min.)	Self-reflection on their expertise with various math skills and growth mindset statements. (SE p. 9)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Translations of Functions		- Chart Paper - Markers (Optional) Sticky Notes		1.1.2 Activity: Gallery Walk – Prepare 8 posters that have the name of the parent function, the equation, a table of values, and its graph. Place posters around the room. Consider placing posters like Exponential Growth and Exponential Decay side-by-side.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may mistake the types of functions based on function features. - Students may mistake horizontal and vertical translations. - Students may mistake the exponential function types.	- Students may not be familiar with each other (or with you). Consider utilizing the Gallery Walk as an opportunity to communicate, establish, and practice classroom norms and discussion norms. Consider providing sentence stems to use during the Gallery Walk to build positive opportunities to engage. - This lesson could easily be divided up in different ways so that while establishing class norms for your class, students are also putting these norms into practice. - The purpose of this lesson is for the student to become familiar with the functions listed in clarification 2 of the benchmark MA.912.F.1.1. Students will continue to interact with most of these functions throughout the course. Full mastery of the parent functions is not an expectation of this lesson. - If time is an issue, then consider assigning 1.1.4 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder Leverage 1.1.3 Guided Practice as a Notice and Wonder to explicitly activate the “noticing” that students expressed during the Gallery Walk. Students are prompted to notice what stayed the “same” to explicitly identify why a parent function remains in that family with translations.	MA.K12.MTR.2.1: Multiple Representations At some point during the lesson, prompt students to discuss the connection between the equation, the table of values, and the graph. Are these three different functions? Are they all the same function? Begin laying the foundation for multiple representations for students using this language.
 Differentiate Instruction	Notice the varying levels of number sense and operations needed for different functions. The comprehension of each type of function can be an indicator of varying proficiency levels for your students to inform student grouping in upcoming lessons.	
	Notice the varying entry points for 1.1.2 Activity. Students are associating locations with graphs and physically interacting, making sense of their “noticing” with a partner verbally, and purposefully making visual connections and using visual cues.	
Consider having students use that same number rubric in the Wrap-Up to label each of the 8 parent functions with their level of comprehension. Some students familiar with linear functions may struggle with cubic functions, indicating a gap in knowledge regarding exponents.		
Lesson Notes:		

1.1.1 Warm-Up: Would You Rather?

Component Overview: Students will complete an introduction activity to get to know other students in the class by finding out what other students would rather do. This activity is designed to be done collaboratively by having students interact with a different student for each piece of the table.



1.1.1 Warm-Up: Would You Rather?

- Complete the chart by moving around the room and asking nine different people "Would you rather ...?" Introduce yourself to each person. Ask each person a different question from your chart. Record their name and their answer.

Answers will vary.

Would you rather ...

fly a kite or swing on a swing?	be a famous inventor or a famous writer?	go skiing or go to a water park?
be able to create a new holiday or create a new sport?	start a colony on another planet or be the leader of a small country on Earth?	own an old pirate ship with crew or a private jet with a pilot?
see a firework display or a circus performance?	be able to talk to land animals or talk to sea animals?	time-travel to the past or the future?



Prior to starting the activity, consider reinforcing any classroom management protocols that you have established as part of this activity. Be as explicit as possible with your instructions. Consider setting a time limit for this activity and reinforcing that time limit to maintain momentum for this lesson.



As an activity modification, consider creating two sides to the room: one side that represents choice 1 and one side that represents choice 2. Narrate the choices for the "would you rather" activity and have students stand on the side they choose.



After completing the activity, consider having students stay with their last partner. Have students introduce their last partner by sharing something the student would rather do. Repeat this for each student pair until all students have had a chance to be introduced to their peers.

1.1.2 Activity: Parent Functions Gallery Walk

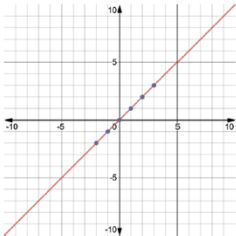
Component Overview: Print and paste (or write and sketch) each function on a poster with its equation, table of values, and graph on a poster with both prompts provided. Place the 8 posters around the room. Students can travel in pairs and choose a poster, or you can set a timer and have each pair spend a certain amount of time before rotating to a specific next poster. Consider labeling each poster with a letter (A-H), and using those letters as anchors. After students have completed all 8 rotations or visited all 8 posters, prompt them to revisit and draw any stars on their favorite “noticing” or “wondering.”



1.1.2 Activity: Parent Functions Gallery Walk

The study of functions is an area of focus in algebra. Functions are classified into families. The **parent function** is the simplest function of each family that preserves the characteristics of the family.

Here are eight parent functions that will be introduced in Algebra 1. After you examine each parent function, answer the questions that follow.

Linear Function																
Parent Function	Table	Graph														
$y = x$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-2</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr></table>	x	y	-2	-2	-1	-1	0	0	1	1	2	2	3	3	
x	y															
-2	-2															
-1	-1															
0	0															
1	1															
2	2															
3	3															
What do you notice about the graph ?	<i>Answers will vary. The graph is a straight line. The graph begins in the bottom left and goes to the upper right.</i>															
What do you notice about the equation ?	<i>Answers will vary. The equation contains only one 'y' and one 'x'. There are no coefficients.</i>															



Potential Instructions (adjust as needed):

“With your partner, you will bring a post-it note and a pen. At your poster, make at least one noticing about the graph and at least one noticing about the equation. There are no right or wrong answers. After the first 3 rotations, you may notice everything you thought of has been said already. If this happens, instead place a check mark on your noticing, OR write something you wonder about the graph or a question you have.”



If you will be doing the activity with multiple classes, consider using post-it notes instead of having students write directly on the poster paper.

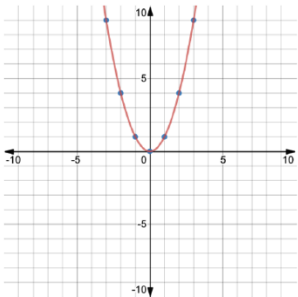


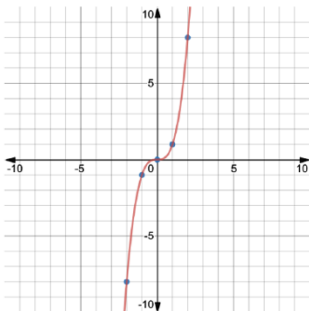
Consider asking prompting questions to guide students to identify unique features that will better position them for success in the following activity.

Example: Linear Function

- What do you notice about the relationship between the ordered pairs on the graph? (reference the table of values if needed)

1.1.2 Activity: Parent Functions Gallery Walk (continued)

Quadratic Function																
Parent Function	Table	Graph														
$y = x^2$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>4</td></tr><tr><td>-1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>9</td></tr></table>	x	y	-2	4	-1	1	0	0	1	1	2	4	3	9	
x	y															
-2	4															
-1	1															
0	0															
1	1															
2	4															
3	9															
What do you notice about the graph ?	<i>Answers will vary. The graph is not a straight line. The graph comes down and then shifts to go back up.</i>															
What do you notice about the equation ?	<i>Answers will vary. The x value is raised to the second power. The equation is not linear.</i>															

Cubic Function																
Parent Function	Table	Graph														
$y = x^3$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-8</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>8</td></tr><tr><td>3</td><td>27</td></tr></table>	x	y	-2	-8	-1	-1	0	0	1	1	2	8	3	27	
x	y															
-2	-8															
-1	-1															
0	0															
1	1															
2	8															
3	27															



Consider what elements of a quadratic function students could identify as unique.

- What do you notice about where the graph “begins” or “ends” on the plane? How might that help you classify this as a quadratic function?
- What about the equation? What unique identifier tells you which type of function it is?
- Also, how does the exponent better inform why it “begins” and “ends” the way it does?

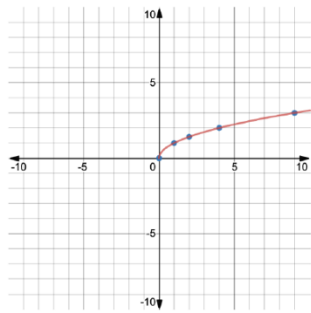


Consider what elements of a cubic function students could identify as unique.

- What do you notice about where the graph “begins” or “ends” on the plane? How might that help you classify this as a cubic function?
- What about the equation? What unique identifier tells you which type of function it is?

1.1.2 Activity: Parent Functions Gallery Walk (continued)

What do you notice about the graph ?	<i>Answers will vary. The graph starts going up and then 'flips' and continues going up. The graph isn't a line and has a 'swoosh' shape.</i>
What do you notice about the equation ?	<i>Answers will vary. The x value is raised to the third power. The equation is not linear.</i>

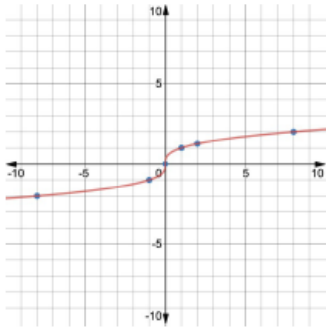
Square Root Function																
Parent Function	Table	Graph														
$y = \sqrt{x}$	<table><tr><th>x</th><th>y</th></tr><tr><td>-1</td><td>not real</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>1.414</td></tr><tr><td>4</td><td>2</td></tr><tr><td>9</td><td>3</td></tr></table>	x	y	-1	not real	0	0	1	1	2	1.414	4	2	9	3	
x	y															
-1	not real															
0	0															
1	1															
2	1.414															
4	2															
9	3															
What do you notice about the graph ?	<i>Answers will vary. The graph starts at the origin and then goes up and to the right in a steep manner. The graph is not a straight line, but instead has a increasing curve.</i>															
What do you notice about the equation ?	<i>Answers will vary. The x value is under the square root symbol. The equation is not linear.</i>															

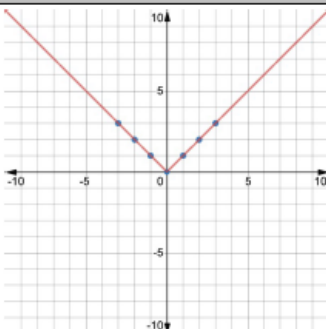


Consider what elements of a square root function students could identify as unique.

- Notice the square root in the equation. What impact does that have on the way the graph looks? Give an example.

1.1.2 Activity: Parent Functions Gallery Walk (continued)

Cube Root Function																
Parent Function	Table	Graph														
$y = \sqrt[3]{x}$	<table><tr><th>x</th><th>y</th></tr><tr><td>-8</td><td>-2</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>1.259</td></tr><tr><td>8</td><td>2</td></tr></table>	x	y	-8	-2	-1	-1	0	0	1	1	2	1.259	8	2	
x	y															
-8	-2															
-1	-1															
0	0															
1	1															
2	1.259															
8	2															
What do you notice about the graph ?	<i>Answers will vary. It begins on the left and shifts towards the origin to 'flip' and continue moving onward. I notice the left and right part of the graph look like reflections of each other.</i>															
What do you notice about the equation ?	<i>Answers will vary. I notice that the equation has x under the radical rather than to a power of 1. I notice the equation is not linear.</i>															

Absolute Value Function																
Parent Function	Table	Graph														
$y = x $	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>2</td></tr><tr><td>-1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr></table>	x	y	-2	2	-1	1	0	0	1	1	2	2	3	3	
x	y															
-2	2															
-1	1															
0	0															
1	1															
2	2															
3	3															



Consider what elements of a cube root function students could identify as unique, particularly to distinguish from a quadratic or square root function.

- Cover up one half of the coordinate plane, either the left or the right. What function does this look like? (square root)
- So, if you uncover that half of the plane, if half is a square root, what do you think the full function might be? (fourth?)
- Why is it a third, or cube, root function? (positive values are reflected in Quadrant 3)
- If you are only given the equation, how could you identify this from other types of functions?

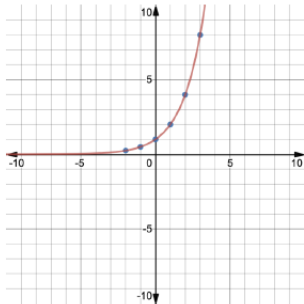


Consider what elements of an absolute value function students could identify as unique.

- Why do you think the graph makes an exact V-shape? Notice anything unique about the table of values that helps? (symmetry)
- What key feature in the equation can serve as a unique identifier?

1.1.2 Activity: Parent Functions Gallery Walk (continued)

What do you notice about the graph ?	<i>Answers may vary. The graph makes a v shape. The graph comes down from the left and then goes up and to the right once it hits the origin.</i>
What do you notice about the equation ?	<i>Answers may vary. The x value is within absolute value bars. This equation is not linear.</i>

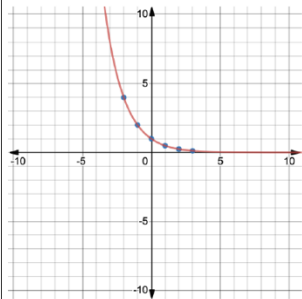
Exponential Function																
Parent Function	Table	Graph														
Exponential Growth Base (2) $y = 2^x$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>$\frac{1}{4}$</td></tr><tr><td>-1</td><td>$\frac{1}{2}$</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>2</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>8</td></tr></table>	x	y	-2	$\frac{1}{4}$	-1	$\frac{1}{2}$	0	1	1	2	2	4	3	8	
x	y															
-2	$\frac{1}{4}$															
-1	$\frac{1}{2}$															
0	1															
1	2															
2	4															
3	8															
What do you notice about the graph ?	<i>Answers will vary. The graph begins with a narrow increase and then increases sharply. The graph goes up slowly and then shoots up quickly.</i>															
What do you notice about the equation ?	<i>Answers will vary. The x-value is in the exponent place. This equation is not linear.</i>															



Consider what elements of an exponential growth function students could identify as unique, particularly the difference between growth and decay.

- Compare this graph to exponential decay. Where does this graph “start” and where does it “end”?
- What does your response tell you about why this function represents growth? (use the table of values to scaffold this question)
- What about the equation? What is the same or different compared to decay?

1.1.2 Activity: Parent Functions Gallery Walk (continued)

Exponential Function																
Parent Function	Table	Graph														
Exponential Decay Base $\left(\frac{1}{2}\right)$ $y = \left(\frac{1}{2}\right)^x$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>4</td></tr><tr><td>-1</td><td>2</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>$\frac{1}{2}$</td></tr><tr><td>2</td><td>$\frac{1}{4}$</td></tr><tr><td>3</td><td>$\frac{1}{8}$</td></tr></table>	x	y	-2	4	-1	2	0	1	1	$\frac{1}{2}$	2	$\frac{1}{4}$	3	$\frac{1}{8}$	
x	y															
-2	4															
-1	2															
0	1															
1	$\frac{1}{2}$															
2	$\frac{1}{4}$															
3	$\frac{1}{8}$															
What do you notice about the graph ?	<i>Answers will vary. The graph starts from the left and decreases quickly. This one looks like the opposite of the exponential growth.</i>															
What do you notice about the equation ?	<i>Answers will vary. The x-value is in the exponent and a fraction is the base. The equation is not linear.</i>															



Consider what elements of an exponential decay function students could identify as unique, particularly the difference between growth and decay.

- Compare this graph to exponential growth. Where does this graph “start” and where does it “end”?
- What does your response tell you about why this function represents decay? (use the table of values to scaffold this question)
- What about the equation? What is the same or different compared to growth?

1.1.3 Guided Practice: Classifying Functions

Component Overview: Students analyze the equations and corresponding graphs of functions to determine the impact of translations of classification. Teachers should monitor student responses in questions 1 and 2. Students apply their understanding by classifying more functions in question 3.



1.1.3 Guided Practice: Classifying Functions

Functions can be transformed by translations, dilations, and reflections.

In this example, the quadratic parent function is translated vertically, horizontally, and reflected over the x -axis.

Quadratic Parent Function	Translated Vertically	Translated Horizontally	Reflected over the x -axis
$y = x^2$	$y = x^2 - 2$	$y = (x - 5)^2$	$y = -(x)^2$

- What stayed the same in each of the equations?
In each equation, the y -value remained on the left hand side of the equation. In each equation, the exponent remained 2.
- What stayed the same in each of the graphs?
In each graph, the size of the parabola (width and height) remained the same.



Notice and Wonder

Utilize a Notice and Wonder routine to position students to formalize specific learning targets. Frame questions 1 – 2 by prompting students to specifically notice how the graph changed in relation to how the equation changed.

- The equation of each function retains an exponent of 2. Students should formalize this exponent as the key feature that can classify a quadratic.
- The “shape” of the graph stays the same. Students should formalize that translated functions still belong to the same family of functions.

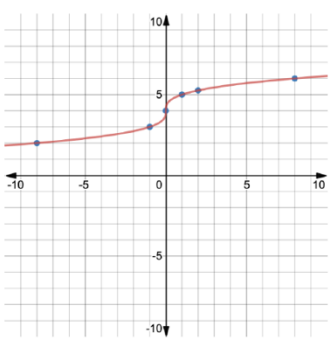
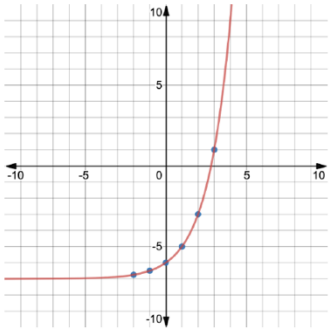


Consider prompting questions to target specific learning outcomes for these questions.

- Is this function still a quadratic function? (For example, “This is still a quadratic function just with a different vertex.”)
- Can you restate what you mean by “move up” but use the word “translate” instead? (“The second function is still a quadratic, but the -2 just means the function was translated 2 units down.”)

1.1.3 Guided Practice: Classifying Functions (continued)

3. Classify each function into its function family based on the characteristics observed for the parent functions.

Function	Function Family
$y = x + 9 $	<i>Absolute Value</i>
$y = -\left(\frac{1}{2}\right)^x$	<i>Exponential Decay</i>
$y = x^3 - 6$	<i>Cubic</i>
	<i>Cube Root</i>
	<i>Exponential Growth</i>



Consider incorporating the Gallery Walk posters as scaffolding for students. Prompt students to answer each question first, and then provide a reason why they chose that function. Students should then visit the poster of their answer to verify their answer.



Consider modeling think-aloud questions to provide students with questions to consider.

- What makes this equation unique?
- What does that feature tell me about the type of function it belongs to?
- Does that have anything to do with what it looks like on the graph?

1.1.4 Your Turn!

Component Overview: Students will independently practice determining the function family given an equation or graph.



1.1.4 Your Turn!

Classify the function family for each equation or graph.

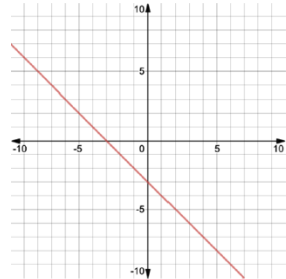
1. $y = (x + 2)^3$ Function Family: Cubic

2. $y = -x - 23$ Function Family: Linear

3. $y = 2^{(x-7)}$ Function Family: Exponential Growth

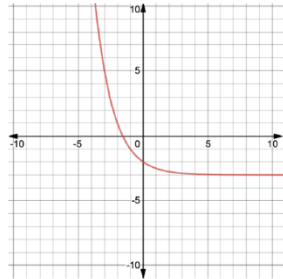
4. $y = (x)^2 + 31$ Function Family: Quadratic

5.



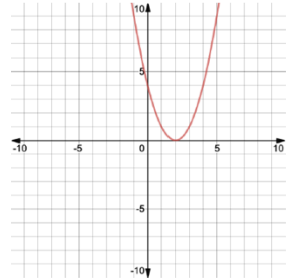
Function Family:
Linear

6.



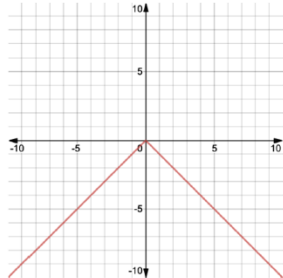
Function Family:
Exponential Decay

7.



Function Family:
Quadratic

8.



Function Family:
Absolute Value



Students have not formally learned about most of the functions presented. Students might struggle most with identifying the unique differences between functions that look similar on a graph (i.e., answering “square root function” to question 6).



Consider prompting questions that identify the unique differences between these functions.

- Question 1: What is the value of the exponent? What does that tell you about the function?
- Question 3: Where is the variable x ? What does that tell you about the type of function?
- Question 6: Note the square root function. What is different about this function?

1.1.5 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection where they rate themselves on their expertise with various math skills and growth mindset statements. This activity is designed to be done independently to get an accurate self-reflection from each student.



1.1.5 Wrap-Up: Reflection

- As you begin your journey through Algebra 1, rate yourself on where you feel you are today on the following skills:

Answers will vary.

Skill	1 – I am just starting. 2 – I have some skills. 3 – I am almost there. 4 – I have it!
I can evaluate expressions using order of operations.	
I can graph points on a coordinate plane.	
I can apply the laws of exponents.	
I can determine whether a relation is a function.	

- Rate yourself based on how you feel about the following statements:

Statement	1 – I struggle with this. 2 – I am ready to grow in this area. 3 – I am almost there. 4 – That's me!
I am willing to try new things.	
I persevere when something is challenging.	
I am a student learner.	
I am a mathematician.	



Notice these statements are not related to this lesson but meant to collect a topical pulse check on a few concepts students will encounter in the course.

Alternatively, consider having students number each activity in this lesson with the same rubric, or number each type of parent function in this lesson with their level of understanding.